Exploring How Scene Homogeneity and Story Continuity Impact Emotion Perception in 360-Degree Virtual Reality Videos

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Abstract

The affective datasets used in immersive 360-degree videos typically rely on short excerpts extracted from longer videos, which are then labeled via self-reporting. However, little is known about the effects of excerpt selection on the fidelity of the emotions that are elicited. This paper outlines an exploratory pilot study on the extraction of one-minute 360-degree VR clips using random segmentation or keyframe-based segmentation configured for visual homogeneity and continuous content. Results show that keyframe-based clips generated stronger responses (i.e., deviating further from neutral affect) and received higher ratings for narrative continuity and event logic; however, none of the observed differences reached the level of statistical significance. The interviews indicated that visual coherence alone was insufficient to ensure comprehension, which also depended on the prior experience of the viewer. Emotional responses varied with viewing direction and perspective, while narration, on-screen text, background music, and personal interest influenced both the emotions experienced by participants and how they were reported. Our findings indicate that scene homogeneity and narrative continuity should support the broader goal of eliciting emotional responses that are consistent, non-neutral, and easy for viewers to report. These insights also provide guidance on clip duration, storyline structure, and emotion-reporting methods, which may improve the design of affective 360-degree video clips and the reliability of emotion labels used to train recognition models.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in ubiquitous and mobile computing**.

Keywords

Emotion Inference, Self Report, Virtual Reality.



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1 Introduction

Immersive 360-degree videos viewed through head-mounted displays (HMDs) are widely used to elicit emotions in laboratory studies. Allowing free visual exploration of an immersive, fully-encompassing space has been shown to strengthen one's emotional responses by enhancing their sense of presence [10, 12]. This has led to the development of several publicly available affective datasets of 360-degree videos with annotations pertaining to valence and arousal [9, 18, 19]. However, many source videos are of long duration (spanning several minutes) and contain heterogeneous visual content and/or discontinuous storylines, which can introduce fatigue, habituation, and inconsistent emotional responses [13]. As a result, many researchers seeking to elicit specific emotions must extract short excerpts from longer videos, a process that depends almost entirely on segment selection and emotion labeling. An excerpt can only be considered useful as a ground truth reference if the emotion it evokes is of sufficient specificity and strength to generate a consistent response across a wide range of viewers.

Narrative theory and the distinctive properties of 360-degree VR both suggest that emotional responses depend largely on segment selection [20]. Narrative research has identified comprehension as a prerequisite for emotional engagement. Viewers who are able to follow the events of a story are more likely to become immersed in its narrative world, thereby intensifying their affective responses [4, 7]. It appears that two properties of an excerpt are of particular importance: scene homogeneity and story continuity. Scene homogeneity is defined as the visual consistency of the scene, and story continuity is defined as the coherence of events within a single narrative [16]. In 360-degree videos, viewers control where they are going and what they are looking at. This makes it difficult to use conventional continuity-editing cues to preserve homogeneity and continuity [16], while increasing the likelihood that viewers will miss details that are crucial to the narrative [11]. Keyframe-based summarization can be configured to favor segments that are both

coherent and representative [1]; however, these methods are generally optimized for browsing, indexing, and retrieval rather than affective impact [21]. At present, there is some debate as to whether a video segment can preserve or strengthen the emotional effect of the source video. We address this gap by comparing keyframe-based and randomly positioned excerpts and examining viewers' reported emotions after watching them in 360-degree VR.

In this exploratory pilot study, we compared the emotional responses elicited by keyframe-based excerpts with those elicited by excerpts selected from random positions. Participants viewed eight one-minute clips spanning all four quadrants of the valence–arousal space through a head-mounted display, and then reported their valence and arousal states using the Self-Assessment Manikin [2]. Keyframe-based clips generated responses that deviated further from neutral affect and received higher ratings for narrative continuity and event logic; however, none of the observed differences reached the level of statistical significance. The interviews revealed that visual coherence alone was insufficient to ensure comprehension, which also depended on the prior experience of the viewer. Emotional responses varied with viewing direction and perspective, while narration, background music, and personal interest influenced both the emotions experienced and how participants reported them. These findings suggest that scene homogeneity and story continuity should be key targets in efforts to elicit emotional responses that are consistent, distinct, and easy for viewers to report. They also provide guidance on clip duration, storyline structure, and emotion-reporting methods, which may improve the design of affective 360-degree video clips and the reliability of emotion labels used to train recognition models.

2 Pilot Study

2.1 Participant

This study used snowball sampling to recruit four participants from an anonymized university (P1 to P4). This sample (one male and three females) included both undergraduate and graduate students and one member of the administrative staff, ranging in age from 20 to 30 years old (mean age 21.0 years, SD = 2.0).

2.2 Device

Emotional responses were elicited by immersing participants in 360-degree VR video clips using a Meta Quest 3 head-mounted display (HMD) with handheld controller. Participants used only the right-hand trigger to provide questionnaire responses while in the VR environment.

2.3 360-Degree VR Video Clips

A pool of 20 publicly available 360-degree videos was compiled from YouTube and the Internet Archive [8]. The affective content of each video was assessed by submitting full-length videos to the vision–language model Gemini 2.5 Flash-Lite [6], which sampled each video at one frame per second and produced an overall valence and arousal rating for each video. These ratings were used to assign each video to one of four quadrants of Russell's valence–arousal space [14].: high valence and high arousal (HVHA; valence ≥ 5 and arousal ≥ 5), high valence and low arousal (HVLA; valence ≥ 5 and arousal < 5), low valence and high arousal (LVHA; valence < 5 and arousal ≥ 5), or low valence and low arousal (LVLA; valence < 5 and arousal < 5). A score of 5 represented the midpoint of the corresponding Self-Assessment Manikin (SAM) scale. Five videos were selected from each quadrant to obtain a roughly balanced stimulus pool. The source videos ranged from 4 to 22 minutes in duration, which was sufficient for the extraction of one-minute clips, a duration that has been proposed as the minimum duration required to elicit noticeable affective responses [15, 17]. We then compared the effects of clips segmented using different segmentation strategies on emotion elicitation.

The effects of scene homogeneity and story continuity on emotional responses were assessed by extracting one one-minute clip from each source video using either random or keyframe-based segmentation. Keyframe-based segmentation was performed using PySceneDetect [5] to identify scene boundaries and then select segments with relatively consistent visual content and event continuity. Random segmentation involved selecting a clip start time without considering scene homogeneity or narrative continuity. Each participant viewed eight one-minute clips spanning of the four valence–arousal quadrants.

2.4 Procedure

The study was conducted in three phases: preparation, video viewing, and a post-study interview.

During the preparation phase (approximately 10 minutes), the researcher introduced the study objectives, explained the experimental procedure, and assisted participants in donning the VR headset. Participants were then familiarized with the VR interface, including how to navigate the virtual environment and complete questionnaires while using point-and-click interactions in VR. They also practiced reporting their emotions using the Self-Assessment Manikin (SAM) scale [2].

During the video viewing phase (roughly 20 minutes), participants watched eight one-minute 360-degree VR clips. The clips were randomly selected from the pool of 20 source videos, with the segmentation method (keyframe-based or random) counterbalanced across participants. The presentation order was randomized, while ensuring that no two consecutive clips were drawn from the same valence–arousal quadrant. Each trial involved watching the video for 60 seconds and then immediately completing an in-VR questionnaire for 60 seconds. Participants rated their perceived emotions in terms of valence and arousal using the 9-point SAM scale (1 = low, 9 = high) and then completed five additional 5-point Likert-style questions assessing perceived scene homogeneity and storyline continuity (Section 2.5).

Following completion of the viewing session, participants engaged in a semi-structured interview lasting roughly 60 minutes. To facilitate recall during the interview, each video clip was replayed while discussing the ratings assigned by the participant. Participants were asked to explain which factors influenced their emotional responses, whether scene transitions or visual changes affected their understanding or attention, and how scene coherence influenced their sense of immersion and emotional engagement.

2.5 Measures

After each video clip, participants completed an in-VR questionnaire. They first rated their affective state using separate 9-point Self-Assessment Manikin (SAM) scales for valence and arousal. Prior to the experiment, both dimensions were explained to the participants using an annotated affect grid. To quantify the magnitude of each response relative to the neutral midpoint of the scale, we calculated the absolute deviations of valence and arousal from 5: $DV = |V - 5|$) and ($DA = |A - 5|$), where (V) and (A) respectively denote the self-reported valence and arousal ratings. Participants then completed five items assessing perceived scene homogeneity and story continuity on a 5-point Likert scale ranging from 1 ("strongly disagree") to 5 ("strongly agree") (Figure 1): Q1, changes in the scene or environment felt like part of a continuous story (narrative continuity); Q2, the storyline remained clear and easy to understand despite changes in the environment (narrative clarity); Q3, the events were connected by a coherent logic (event logic); Q4, visually consistent or unified elements made the scene more engaging (consistency appeal); and Q5, visually diverse or heterogeneous elements were distracting (diversity distraction).

2.6 Data Analysis

Responses were grouped according to segmentation strategy: keyframe-based and random segmentation. We used significance tests to compare the two segmentation methods in terms of DV, DA, and scene-homogeneity and storyline-continuity ratings.

All semi-structured interviews were audio-recorded and transcribed verbatim. A qualitative dataset comprising post-study interview transcripts was analyzed using thematic analysis [3]. Two experienced researchers independently reviewed and coded the data through multiple iterative rounds to identify themes related to the effects of scene homogeneity and story continuity on narrative comprehension, emotional engagement, and the ease of self-reporting the elicited emotion. They then discussed discrepancies in their coding results until reaching consensus. The agreed-upon codes were subsequently grouped into three higher-level themes.

2.7 Results

2.7.1 Quantitative Results. Clips generated via keyframe-based segmentation elicited stronger responses than did randomly segmented clips, as indicated by the degree of deviation from the neutral midpoint of 5. The mean DV of keyframe segmentation (1.75) was 8.0% higher than that of random segmentation (1.62) (p = 0.76), while the mean DA of keyframe segmentation (1.94) was 7.2% higher than that of random segmentation (1.81) (p = 0.76). These results suggest that keyframe-based segmentation may produce affective responses that deviate more strongly from neutral [4, 7].

Figure 1 summarizes the mean ratings for narrative continuity (Q1), narrative clarity (Q2), event logic (Q3), consistency appeal (Q4), and diversity distraction (Q5) under the keyframe-based and random segmentation conditions. The mean rating for narrative continuity (Q1) in videos based on keyframe segmentation was 14.8% higher than those based on random segmentation (p = 0.17) and the mean rating for event logic (Q3) was 16.0% higher (p = 0.23). The rating distributions for narrative clarity (Q2), consistency appeal (Q3), and diversity distraction (Q5) were nearly identical between the two segmentation methods (p = 0.82, 0.89, and 1.00, respectively).

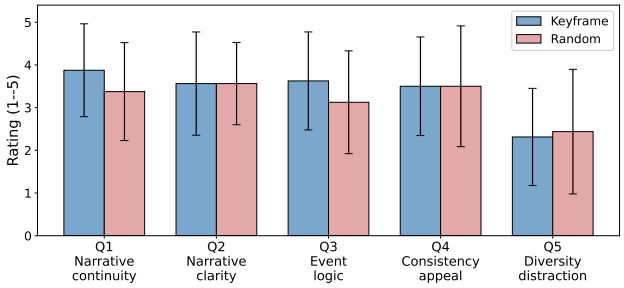


Figure 1: Mean and standard deviation of ratings assessing perceived scene homogeneity and story continuity.

2.7.2 Qualitative Results.

2.7.2.1 Narrative Meaning-Making Through Visual Coherence and Prior Experience. Participants found it easier to understand the intended storyline when the clip maintained a consistent visual setting as well as a coherent sequence of related events. Keyframe-based segmentation appeared to be more effective in preserving a stable context, a recognizable topic, and visually connected actions. This consistency eliminated the need to repeatedly reinterpret the environment or determine how each new scene related to the previous one. Instead, the clip could be experienced as a unified event, with little confusion interfering with engagement. P1 made the following comment: *"I found the underwater shark scene easier to understand then the previous videos, Horror Roller Coaster and Brazil's Disappearing Wild Jaguar, because it stayed within a consistent ocean setting. The roller coaster video had more scene transitions, and the jungle video mixed wild animals with human narration. This clip focused on sharks, so the theme was clear. The sharks swimming around and opening their mouths matched the ocean environment, making the video feel coherent and engaging."*

Prior experience supported the interpretation of the narrative and also shaped emotional responses. When participants associated the clip with a personally meaningful or emotionally salient real-life event, they anticipated similar outcomes and responded more strongly. In such cases, the emotional effect of the video was not produced solely by the current visual content. Instead, memories and expectations from previous experiences amplified feelings of fear, tension, or panic. Once participants understood or constructed a plausible storyline, visually diverse or unexpected elements were less likely to be perceived as distracting. Instead, viewers interpreted these elements in relation to the genre and narrative context. For example, strange objects, sudden appearances, and rapid visual changes could be accepted as appropriate in a horror-themed roller coaster experience. This suggests that narrative framing can provide a meaningful structure conducive to the integration of heterogeneous visual elements, allowing them to reinforce rather than disrupt the intended emotional atmosphere. P4 made the following comment: *"While watching the Horror Roller Coaster video, I was not distracted by the varied visual elements. The clown effects and floating mouths and teeth felt connected to the horror storyline. I*

realized that it was a horror-themed amusement ride, so I expected strange objects to appear. The rapidly changing visuals felt logical and coherent rather than confusing, and they helped to create the intended atmosphere."

2.7.2.2 Narrative Guidance Across Viewing Perspectives for Emotional Engagement. Even when keyframe-based segmentation preserves continuity over time, different viewers are likely to encounter different environmental or social cues. As a result, different users often experience different emotional responses within the same scene, depending on where they look. Thus, any added detail across the VR environment could act as a distraction or add emotional complexity. Participants' responses indicate that a dynamic narrative anchor, such as a person, object, or unfolding event, can guide the attention of the viewer and clarify the progression of the video. Viewers should not be left to search the environment for meaning. They should be provided a focal point that acts as a continuous cue connecting successive moments and viewing positions. This can make the storyline easier to follow while also supporting engagement. P2 made the following comment: *"The sixth video [Isle of Jaws] was easy to understand. When the ball containing the camera was released, my attention naturally followed as it descended into the water. I just followed the scene it captured, so the main focus remained clear. The storyline also felt logical because the above-water and underwater scenes were connected through the same camera and the same goal of searching for and filming sharks."*

The viewing perspective also shaped how participants experienced the clips. A first-person perspective could strengthen a sense of presence by positioning viewers within the unfolding action, allowing them to experience events as though they were experiencing them directly. However, the first-person perspective is not inherently engaging. The viewer must have an interest in the subject matter for it to be effective. P4 made the following comment: *"Across these videos, the first-person perspective had the strongest effect on me when I was interested in the scene. In engaging clips, such as walking through the streets of Jerusalem in the 'Via Dolorosa in Jerusalem' video, riding a roller coaster in the 'Horror Roller Coaster' video, or exploring underwater in the 'Titanic Wreck' video, the first-person perspective increased curiosity and my sense of involvement. However, when the content was less interesting, the first-person view felt restrictive and boring. Overall, the first-person perspective enhanced my sense of immersion only when the scene itself captured my attention."* The third-person perspective is effective in maintaining narrative comprehensibility while creating a sense of emotional distance. When viewers observed events through explanatory narration or documentary-style presentation, they could understand what was happening and recognize the intended emotional tone. Nevertheless, third-person presentations are less effective in involving the viewer in the scene or eliciting emotions. P3 made the following comment: *"Visual consistency helped me focus on the scene and my emotions. Even when later videos contained more varied visuals, I could still understand them, but they felt more like news reports and created greater emotional distance. Third-person narration made it harder to place myself in the scene, whereas first-person videos, such as flying an aircraft or exploring underwater, made me feel more present and directly involved."*

2.7.2.3 Verbal, Auditory, and Personal Cues Support Narrative Interpretation and Emotional Engagement. The responses of participants suggest that narration can function as both a narrative guide and an emotional cue. By explaining the purpose of an activity and directing attention toward key events, narration can clarify for viewers the unfolding storyline and give them a sense of what may happen next. Anticipation allows emotions such as curiosity, tension, and surprise to develop through the unfolding narrative rather than in response to isolated visual events. Background music also shaped the interpretation of the scenes and the emotions triggered by the experience. When the storyline was ambiguous, music could provide affective cues suggestive of whether a scene should be interpreted as tense, mysterious, pleasant, or unsettling. Nonetheless, unlike narration or text, music cannot always clarify what is happening. When paired with unclear visual content, unsettling music could reinforce a negative atmosphere while simultaneously increasing confusion and discomfort. P3 made the following comment: *"While watching the fifth video, I was unsure what was happening, and the strange background music made the experience feel eerie. I felt uncomfortable. Because I could not understand the purpose of the scene, the unsettling sound effects confused me even more. Although the music did not clarify the storyline, it suggested that something strange or disturbing was happening, which intensified my negative emotional response."* The personal preferences and interests of the viewer can also shape the interpretation of the video and emotional engagement. When the content aligned with an existing interest, viewers paid greater attention to important elements, easily understood their significance, and experienced stronger emotions. Personal interest acted as an interpretive and affective lens, amplifying the influence of both visual and auditory cues. P2 made the following comment: *"I felt very happy and excited because I like animals and found the deer cute. Seeing so many of them eating together made me want to reach out and touch them, and the background music added to the pleasant feeling. I was slightly surprised by the number of deer in the scene, but my main emotional response came from my personal interest in animals."*

3 Discussion

3.1 Emotion Perception and the Limits of Self-Report

The dimensional ratings should be interpreted cautiously because participants did not find valence and arousal equally easy to identify and report. Several participants were unsure whether the 360-degree experience felt pleasant or unpleasant. Rather than reporting a clearly perceived emotional tone, some inferred valence by elimination, reasoning that the absence of a positive feeling implied a negative response. Although arousal was somewhat easier to rate, participants were not always able to distinguish between emotional arousal elicited by the video and heightened alertness caused by the novelty of the VR experience. Self-report remains the standard, and often the only practical, method for measuring felt emotion [2]. Nevertheless, a single valence–arousal rating can provide only a coarse summary of an experience that may be complex, mixed, or uncertain. This limitation may be particularly important in 360-degree VR, where emotional experience depends partly on where the viewer chooses to direct their attention. Unlike conventional

two-dimensional video, in which the editor controls the camera, 360-degree VR allows different viewers to encounter different emotional cues within the same clip [11]. These differences in viewing direction, together with the possibility of experiencing multiple emotions within a single clip, may have increased variation among participants and made differences between the two segmentation conditions more difficult to detect.

3.2 Design Implications for Building Affective 360-Degree Video Datasets to Facilitate Reliable Affective Data Collection

Taken together, these findings suggest that constructing affective VR stimuli requires attention to more than the method used to segment a clip. First, because different viewers may encounter different emotional cues within the same clip, a single label may not capture the full range of responses it elicits. Affective datasets may therefore benefit from view-resolved or time-resolved annotations that record how viewers' emotions change across viewing directions and over time [19]. Second, coherence-oriented keyframe segmentation addresses visual consistency and narrative continuity, but emotional responses may also be influenced by audio, on-screen text, salient focal objects, personal interest, and the underlying structure of the source video. Source videos should therefore be screened for narrative cues, such as narration, subtitles, or clearly identifiable focal events, that make short excerpts easier to understand. Audio and on-screen text should also be treated as relevant analytical features rather than excluded from consideration. Third, selecting clips that align with viewers' interests may increase engagement and produce clearer emotional responses.

4 Conclusion

This paper presents an exploratory pilot study examining the relationships among scene homogeneity, story continuity, and emotion perception in one-minute 360-degree VR clips selected using either keyframe-based or random segmentation. Post-study interviews indicated that the emotional experiences of the participants depended on more than visual coherence. Comprehension and emotional responses were also shaped by prior experience, viewing direction and perspective, narration, background music, and personal interests. The selection and design of affective VR clips should therefore consider scene homogeneity and story continuity alongside clip duration, storyline structure, narrative and auditory cues, viewing perspective, the characteristics of the source video, and the limitations of emotion self-report. Attention to these factors may support the development of affective 360-degree datasets that elicit more consistent responses, yield more reliable self-reported ground-truth labels, and improve the training of emotion-recognition models [19].

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